mTeSS-X: Scaling the TeSS for federated training portals through multi-spacing and exchange

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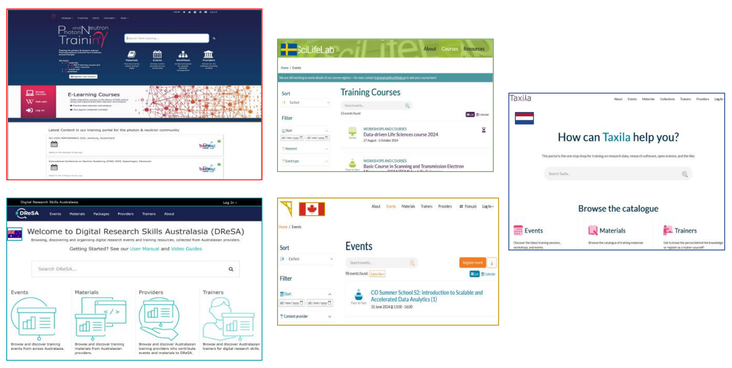
Description

ELIXIR's open-source Training e-Support System (**TeSS**) has served the life sciences community as a one-stop-shop for trainers and trainees to access training materials and events in bioinformatics, data science, and computational biology. The TeSS platform also forms the basis of many independent training instances, however, content sharing and metadata exchange between these instances are not supported. The **OSCARS mTeSS-X** project builds upon the TeSS platform to address this fragmentation of training resources across scientific domains, institutes, and portals and provide a system that allows sharing of nominated content and exchange of metadata between instances. mTeSS-X introduces the concept of "**spaces**", in which communities can have their own separate, self-managed catalogue of training content within a "**hub**" instance of TeSS, and selected content can be **shared** across spaces for the benefit of the wider training community. These enhancements to TeSS will facilitate the integration of additional research infrastructures and science clusters, fostering a more interconnected training landscape.

Training catalogues supported by ELIXIR TeSS



- **Digital Research Skills Australasia** (https://dresa.org.au/)
- **Photon and Neutron** (PaN) training (https://pan-training.eu/)
- **Taxila** Open science training in the Netherlands (https://taxila.nl)
- **SciLifeLab** (https://training.scilifelab.se/)

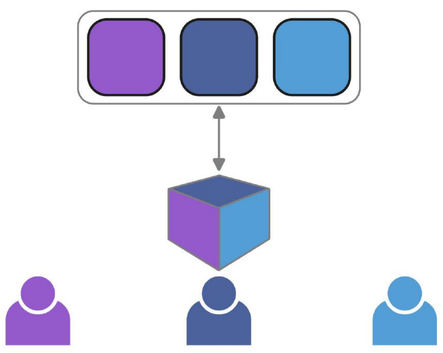
Challenge: Independent Fragmented Training Portals

- Catalogues supported by TeSS are currently:
  - **Completely independant**
  - **Operate as siloed systems**
  - **Do not support content exchange**
  - **Lack shared metadata**
- No support for cross-discipline training resource exchange
- Duplication of operational effort across platforms

mTeSS-X: A federation of training catalogues

Enabling multi-space views and selected content exchange

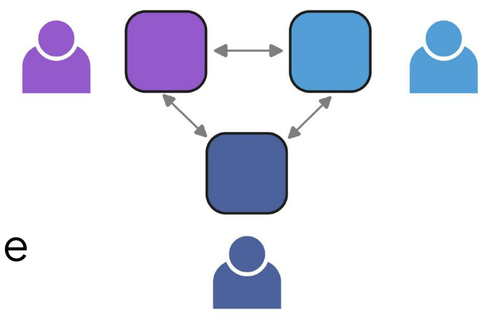
Multi-space catalogue



- Catalogues pooled into a **shared** TeSS instance supporting distinct community-specific "views" or "**spaces**" with:
  - Unique **branding** and identity
  - **Targeted** content selection
- Shared catalogue management

Exchange between catalogues

- Automatic **exchange** of nominated **content** between the spaces in a **TeSS hub**
- Done via a dedicated interface for **metadata harvesting**



Scientific Impact

- **Improved Findability & Accessibility:** Federated catalogue system enables unified discovery across disciplines
- **Increased Reusability:** Content can be reused across domains and portals, promoting FAIR training principles
- **Enhanced Sustainability:** Reduced duplication and operational overhead through shared infrastructure
- **Scalability:** Framework supports future inclusion of more science clusters, communities, and RIs

Join the mTeSS-X Community

The mTeSS-X Club is an open community to help its users and gather feedback for continuous development of the mTeSS-X.

- Bi-weekly Zoom meetings
- Quarterly focus group meetings
- Slack channel: **#tess-club** (elixir-europe.slack.com)

Contact:
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